EXP 801 DS Wheat Heat Drought

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Li-6800 STOMATAL CONDUCTANCE MODEL PARAMETER ESTIMATION - LIGHT OFF TO LIGHT ON (INCREASE)

- Using the final model fitted to the data, obtain the parameters of interest
- These include:
- k_i, being the time constant for gs to increase to Gsmax
- lambda, being the lag in the response time of gs to a step increase in light
- g0, being the starting value of the gs response (intercept/lower asymptote)
- g1, being the upper asymptote of the gs response (this will be used to calculate iWUE)
- Slmax, a derived trait from McAusland paper
-

Load the libraries

```
library(knitr)
#knitr::spin("Exp801.li6800.stomatalCondParamEstimation_i.v1.r")
library(ggplot2)
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300,
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, fig.path = "figures/increase/")
library(dae)
library(dplyr)
library(brms) ##package used to fit bayesian regression model using Stan
library(tidyr)
library(emmeans)
library(tidybayes)
```

Set up constants and files

```
options(width=90)
load("../data/stomatalCondModel_i.v2.Rdata")
saveFile <- "../data/stomatalCondParamEstimation_i.v1.Rdata"
```

Have a look at the final model summary and store fixed effects

```
summary(fit_on_final)
```

```
Warning: There were 21 divergent transitions after warmup. Increasing adapt_delta above
0.99 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - tmin)/k) + 1))
         g0 ~ 1 + Room_Temperature * Watering + (1 | Rep)
         g1 ~ 1 + Room_Temperature * Watering + (1 | Rep)
         k ~ 1 + Room_Temperature
         lambda ~ 1
   Data: li6800.on.dat (Number of observations: 1462)
  Draws: 6 chains, each with iter = 10000; warmup = 6000; thin = 1;
         total post-warmup draws = 24000
```

```
Multilevel Hyperparameters:
~Rep (Number of levels: 3)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.01      0.01     0.00     0.05 1.00     7756    10949
sd(g1_Intercept)     0.02      0.02     0.00     0.08 1.00     5627     7696

Regression Coefficients:
                                        Estimate Est.Error l-95% CI u-95% CI Rhat
g0_Intercept                                0.01      0.01    -0.01     0.02 1.00
g0_Room_TemperatureCER8_Amb                -0.02      0.01    -0.03    -0.01 1.00
g0_WateringUnlim                           -0.01      0.01    -0.02     0.00 1.00
g0_Room_TemperatureCER8_Amb:WateringUnlim    0.02      0.01     0.00     0.03 1.00
g1_Intercept                                0.34      0.01     0.31     0.37 1.00
g1_Room_TemperatureCER8_Amb                -0.13      0.00    -0.14    -0.12 1.00
g1_WateringUnlim                            0.04      0.00     0.03     0.04 1.00
g1_Room_TemperatureCER8_Amb:WateringUnlim    0.10      0.01     0.09     0.11 1.00
k_Intercept                                 3.81      0.26     3.33     4.33 1.00
k_Room_TemperatureCER8_Amb                  2.26      0.26     1.75     2.77 1.00
lambda_Intercept                            2.91      0.43     2.04     3.72 1.00
                                        Bulk_ESS Tail_ESS
g0_Intercept                               17601    16955
g0_Room_TemperatureCER8_Amb                19731    17878
g0_WateringUnlim                           23298    18061
g0_Room_TemperatureCER8_Amb:WateringUnlim  23462    19473
g1_Intercept                                8331     7204
g1_Room_TemperatureCER8_Amb                18562    18522
g1_WateringUnlim                           18714    17275
g1_Room_TemperatureCER8_Amb:WateringUnlim  16922    17677
k_Intercept                                14938    14344
k_Room_TemperatureCER8_Amb                 20443    15960
lambda_Intercept                           14254    12070

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.05      0.00     0.05     0.05 1.00    27461    17232

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Build a dataframe of model parameters

```
li6800.i.param <- expand.grid(Room_Temperature = levels(li6800.on.dat$Room_Temperature),
                              Watering = levels(li6800.on.dat$Watering))
```

Use the emmeans package to get out the expected parameter estimates, along with HPD (credible intervals)

```
g0.emmeans <- as.data.frame(emmeans(fit_on_final,
                                    ~ Room_Temperature * Watering,
                                    nlpar = "g0"))
g0.emmeans <- g0.emmeans %>% rename_with(~paste0("g0_", .x),
                                         -all_of(c("Room_Temperature",
```

```
                                                  "Watering")))
g0.emmeans <- within(g0.emmeans, {
  g0_tabquant <- paste0(round(g0_emmean, 3), " [",
                        round(g0_lower.HPD, 3), ", ",
                        round(g0_upper.HPD, 3), "]")

})
li6800.i.param <- left_join(li6800.i.param, g0.emmeans)
```

```
Joining with `by = join_by(Room_Temperature, Watering)`
```

```
g1.emmeans <- as.data.frame(emmeans(fit_on_final,
                                    ~ Room_Temperature * Watering,
                                    nlpar = "g1"))
g1.emmeans <- g1.emmeans %>% rename_with(~paste0("g1_", .x),
                                         -all_of(c("Room_Temperature",
                                                   "Watering")))
g1.emmeans <- within(g1.emmeans, {
  g1_tabquant <- paste0(round(g1_emmean, 3), " [",
                        round(g1_lower.HPD, 3), ", ",
                        round(g1_upper.HPD, 3), "]")

})
li6800.i.param <- left_join(li6800.i.param, g1.emmeans)
```

```
Joining with `by = join_by(Room_Temperature, Watering)`
```

```
k.emmeans <- as.data.frame(emmeans(fit_on_final,
                                   ~ Room_Temperature,
                                   nlpar = "k"))
k.emmeans <- k.emmeans %>% rename_with(~paste0("k_", .x),
                                       -all_of(c("Room_Temperature")))
k.emmeans <- within(k.emmeans, {
  k_tabquant <- paste0(round(k_emmean, 3), " [",
                       round(k_lower.HPD, 3), ", ",
                       round(k_upper.HPD, 3), "]")

})
li6800.i.param <- left_join(li6800.i.param, k.emmeans)
```

```
Joining with `by = join_by(Room_Temperature)`
```

```
lambda.emmeans <- as.data.frame(emmeans(fit_on_final,
                                  ~ 1,
                                  nlpar = "lambda"))
lambda.emmeans <- lambda.emmeans %>% rename_with(~paste0("lambda_", .x))
lambda.emmeans <- within(lambda.emmeans, {
  lambda_tabquant <- paste0(round(lambda_emmean, 3), " [",
                            round(lambda_lower.HPD, 3), ", ",
                            round(lambda_upper.HPD, 3), "]")

})
li6800.i.param <- cbind(li6800.i.param, data.frame(lambda.emmeans[,2:5]))
```

```
Warning in data.frame(..., check.names = FALSE): row names were found from a short
variable and have been discarded
```

Using these parameters, calculate Slmax

- Slmax looks to be reported in the units mmol m^-2 s^-2
- This involves multiplying g1 and g0 by 1000 and also k by 60
- I can calculate this using the posterior draws for each parameter and then combine them together

```
g0_draws <- emmeans(fit_on_final,
                    ~ Room_Temperature * Watering,
                    nlpar = "g0") %>%
  gather_emmeans_draws()

g1_draws <- emmeans(fit_on_final,
                    ~ Room_Temperature * Watering,
                    nlpar = "g1") %>%
  gather_emmeans_draws()

k_draws <- emmeans(fit_on_final,
                   ~ Room_Temperature,
                   nlpar = "k") %>%
  gather_emmeans_draws()

combined_draws <- g0_draws %>%
  rename(g0 = .value) %>%
  left_join(g1_draws %>% rename(g1 = .value),
            by = c(".draw", "Room_Temperature", "Watering")) %>%
  left_join(k_draws %>% rename(k = .value),
            by = c(".draw", "Room_Temperature"))

combined_draws <- combined_draws %>%
  mutate(Slmax = (g1*1000 - g0*1000) / (exp(1) * k*60))

summary_slopes <- combined_draws %>%
  group_by(Room_Temperature, Watering) %>%
  median_hdi(Slmax) %>%
  rename(Slmax_emmean = Slmax) %>%
  rename(Slmax_lower.HPD = .lower) %>%
  rename(Slmax_upper.HPD = .upper)

summary_slopes <- within(summary_slopes, {
  Slmax_tabquant <- paste0(round(Slmax_emmean, 3), " [",
                           round(Slmax_lower.HPD, 3), ", ",
                           round(Slmax_upper.HPD, 3), "]")

})

li6800.i.param <- left_join(li6800.i.param,
                            summary_slopes %>% select(Room_Temperature,
                                                      Watering,
                                                      Slmax_emmean,
                                                      Slmax_lower.HPD,
                                                      Slmax_upper.HPD,
                                                      Slmax_tabquant))
```

```
Joining with `by = join_by(Room_Temperature, Watering)`
```

```
li6800.i.param
```

```
  Room_Temperature Watering    g0_emmean g0_lower.HPD g0_upper.HPD            g0_tabquant
1        CER7_Heat      Lim  0.005715822  -0.01067725 0.0204419120   0.006 [-0.011, 0.02]
2         CER8_Amb      Lim -0.015632347  -0.03456335 0.0004020862     -0.016 [-0.035, 0]
3        CER7_Heat    Unlim -0.002984951  -0.02342396 0.0153516116 -0.003 [-0.023, 0.015]
4         CER8_Amb    Unlim -0.004560178  -0.02633910 0.0152058786 -0.005 [-0.026, 0.015]
  g1_emmean g1_lower.HPD g1_upper.HPD          g1_tabquant k_emmean k_lower.HPD
1 0.3368347    0.3063949    0.3663453 0.337 [0.306, 0.366] 3.809266    3.328434
2 0.2094931    0.1797629    0.2394620  0.209 [0.18, 0.239] 6.063799    5.437185
3 0.3723058    0.3430099    0.4029144 0.372 [0.343, 0.403] 3.809266    3.328434
4 0.3422122    0.3103194    0.3705435  0.342 [0.31, 0.371] 6.063799    5.437185
  k_upper.HPD           k_tabquant lambda_emmean lambda_lower.HPD lambda_upper.HPD
1    4.327638 3.809 [3.328, 4.328]       2.92598         2.053094         3.730533
2    6.651428 6.064 [5.437, 6.651]       2.92598         2.053094         3.730533
3    4.327638 3.809 [3.328, 4.328]       2.92598         2.053094         3.730533
4    6.651428 6.064 [5.437, 6.651]       2.92598         2.053094         3.730533
       lambda_tabquant Slmax_emmean Slmax_lower.HPD Slmax_upper.HPD       Slmax_tabquant
1 2.926 [2.053, 3.731]    0.5335851       0.4546119       0.6180032 0.534 [0.455, 0.618]
2 2.926 [2.053, 3.731]    0.2276378       0.1904763       0.2709143  0.228 [0.19, 0.271]
3 2.926 [2.053, 3.731]    0.6045666       0.5244643       0.6958533 0.605 [0.524, 0.696]
4 2.926 [2.053, 3.731]    0.3506607       0.3047004       0.4000952   0.351 [0.305, 0.4]
```

Format this in a nicer way for export to Excel

```
li6800.i.export <- li6800.i.param %>% select(Room_Temperature, Watering,
                                             grep("tabquant", colnames(.))) %>%
  pivot_longer(cols = 3:ncol(.),
               names_to = "params",
               values_to = "values") %>%
  pivot_wider(names_from = c("Room_Temperature", "Watering"),
              values_from = "values")
li6800.i.export
```

```
# A tibble: 5 × 5
  params          CER7_Heat_Lim        CER8_Amb_Lim         CER7_Heat_Unlim CER8_Amb_Unlim
  <chr>           <chr>                <chr>                <chr>           <chr>
1 g0_tabquant     0.006 [-0.011, 0.02] -0.016 [-0.035, 0]   -0.003 [-0.023... -0.005 [-0.02...
2 g1_tabquant     0.337 [0.306, 0.366] 0.209 [0.18, 0.239]  0.372 [0.343, ... 0.342 [0.31, ...
3 k_tabquant      3.809 [3.328, 4.328] 6.064 [5.437, 6.651] 3.809 [3.328, ... 6.064 [5.437,...
4 lambda_tabquant 2.926 [2.053, 3.731] 2.926 [2.053, 3.731] 2.926 [2.053, ... 2.926 [2.053,...
5 Slmax_tabquant  0.534 [0.455, 0.618] 0.228 [0.19, 0.271]  0.605 [0.524, ... 0.351 [0.305,...
```

Save data and export

```
save.image(saveFile)
write.csv(li6800.i.export, "../data/li6800.i.param.export.csv", row.names = FALSE)
```